Frequency Response of Transistors

Ginanjar S. Adi

Objectives

1. Understand the concept of decibels, and be able to accurately read a logarithmic plot
2. Become acquainted with the frequency response of a BJT and FET amplifier
3. Be able to normalize a frequency plot, establish the dB plot, and find the cutoff frequencies and bandwidth
4. As a first step in clarifying the relationship between the variables of a logarithmic function, consider the following mathematical equations:

Introduction

- For the amplifier, it was a frequency that normally permitted ignoring the effects of the capacitive elements, reducing the analysis to one that included only resistive elements and sources of the independent and controlled variety.
- We will now investigate the frequency effects introduced by the larger capacitive elements of the network at low frequencies and the smaller capacitive elements of the active device at high frequencies.
- Because the analysis will extend through a wide frequency range, the logarithmic scale will be defined and used throughout the analysis.
- In addition, because industry typically uses a decibel scale on its frequency plots, the concept of the decibel is introduced in some detail.

Logarithms

- As a first step in clarifying the relationship between the variables of a logarithmic function, consider the following mathematical equations:

$$a = b^x, \qquad x = \log_b a$$

$$a = b^x = (10)^2 = 100$$

$$x = \log_b a = \log_{10} 100 = 2$$

Logarithms

- For the electrical/electronics industry and in fact for the vast majority of scientific research, the base in the logarithmic equation is chosen as either 10 or the number *e* = 2.71828..

Common logarithm: $x = \log_{10} a$

Natural logarithm: $y = \log_e a$

- On scientific calculators, the common logarithm is typically denoted by the **log** key and the natural logarithm by the **ln** key.

$$\log_e a = 2.3 \log_{10} a$$

Properties of Logarithms

$$\log_{10} 1 = 0$$

$$\log_{10} \frac{1}{b} = -\log_{10} b$$

$$\log_{10} ab = \log_{10} a + \log_{10} b$$

$$\log_{10} \frac{a}{b} = \log_{10} a - \log_{10} b$$

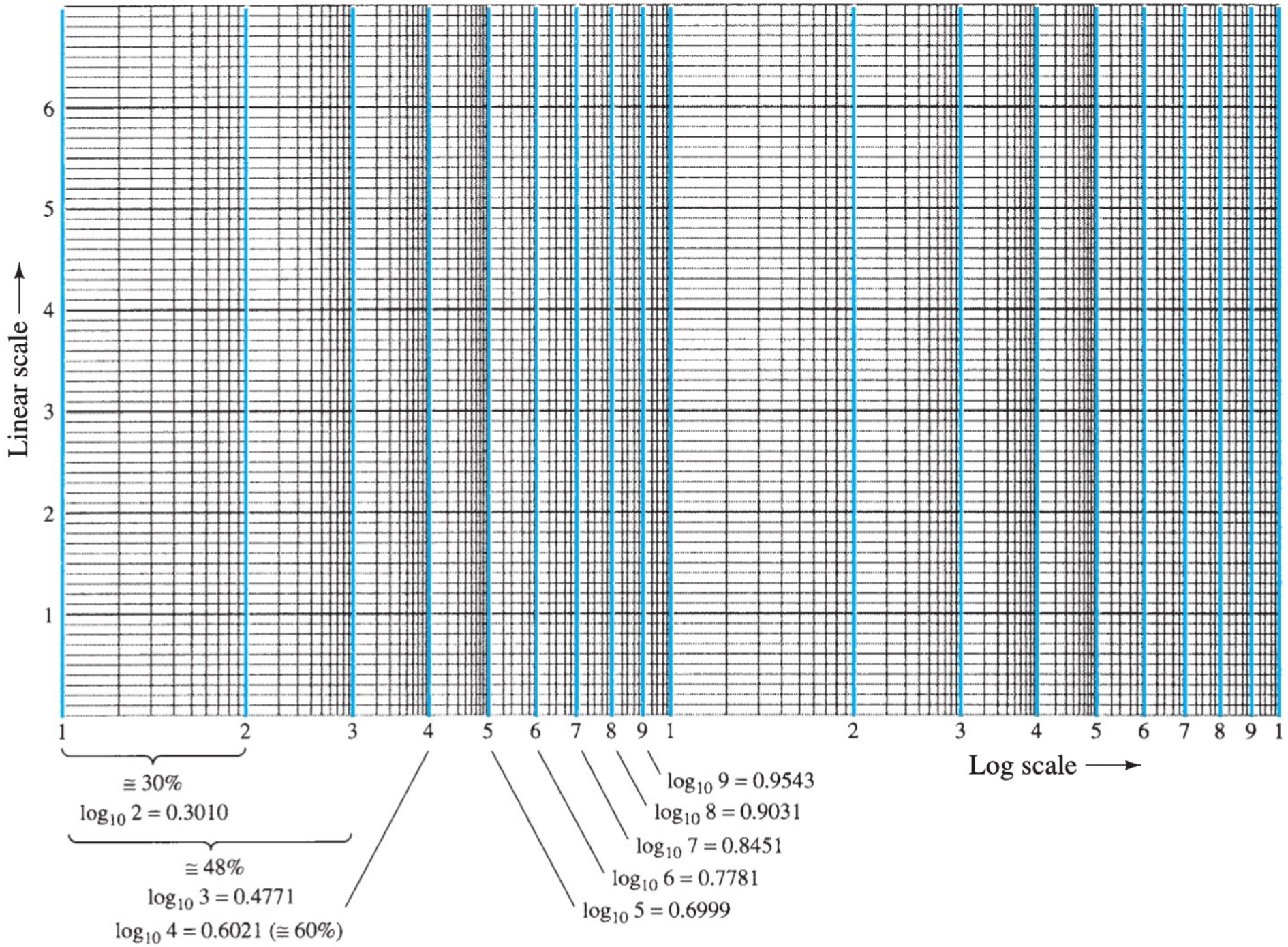
- Exercise:

$\log_{10} 1600.$
$\log_{10} 8000.$

$\log_{10} 0.5.$

$\log_{10} \frac{4000}{250}.$

Semilog Graph Paper



Decibels

- The term *decibel* has its origin in the fact that power and audio levels are related on a logarithmic basis.
- That is, an increase in power level from, say, 4 W to 16 W does not result in an audio level increase by a factor of 16/4 = 4, but by a factor of 2, as derived from the power of 4 in the following manner: $(4)^2 = 16$.
- For a change of 4 W to 64 W, the audio level will increase by a factor of 3 because $(4)^3 = 64$. In logarithmic form, the relationship can be written as $\log_4 64 = 3$.

Power Levels

- The term *bel* is derived from the surname of Alexander Graham Bell. For standardization, the bel (B) is defined by the following equation relating two power levels, P_1 and P_2:

$$G = \log_{10} \frac{P_2}{P_1} \quad \text{bel}$$

- It was found, however, that the bel was too large a unit of measurement for practical purposes, so the decibel (dB) is defined such that 10 decibels = 1 bel. Therefore,

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} \quad \text{dB}$$

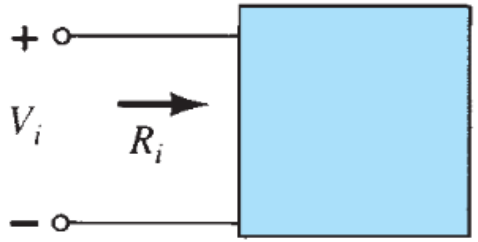
Cont…

- The terminal rating of electronic communication equipment (amplifiers, microphones, etc.) is commonly in decibels.
- For a specified terminal (output) power (*P*2) there must be a reference power level (*P*1).
- The reference level is generally accepted to be 1 mW. The resistance associated with the 1-mW power level is 600 Ohm, chosen because it is the characteristic impedance of audio transmission lines.
- When the 1-mW level is employed as the reference level, the decibel symbol frequently appears as dBm. In equation form,

$$G_{\text{dBm}} = 10 \log_{10} \frac{P_2}{1\ \text{mW}}\bigg|_{600\ \Omega} \quad \text{dBm}$$

Cont…

- There exists a second equation for decibels that is applied frequently. It can be best de- scribed through the system of Fig. 9.5. For *Vi* equal to some value V_1, $P_1 = V_1^2 > R_i$, where R_i is the input resistance of the system of Fig:



- If V_i should be increased (or decreased) to some other level, V_2 then $P_2 = V_2^2 > R_i$. If we substitute into prev Eq. to determine the resulting difference in decibels between the power levels, we obtain

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{V_2^2/R_i}{V_1^2/R_i} = 10 \log_{10} \left(\frac{V_2}{V_1} \right)^2$$

$$\boxed{G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1}} \quad \text{dB}$$

Cascaded Stages

- One of the advantages of the logarithmic relationship is the manner in which it can be applied to cascaded stages. For example, the magnitude of the overall voltage gain of a cascaded system is given by

$$|A_{v_T}| = |A_{v_1}| \cdot |A_{v_2}| \cdot |A_{v_3}| \cdots |A_{v_n}|$$

- Applying the proper logarithmic relationship results in

$$G_v = 20 \log_{10} |A_{v_T}| = 20 \log_{10} |A_{v_1}| + 20 \log_{10} |A_{v_2}| + 20 \log_{10} |A_{v_3}| + \cdots + 20 \log_{10} |A_{v_n}| \qquad \text{(db)}$$

$$\boxed{G_{\text{dB}_T} = G_{\text{dB}_1} + G_{\text{dB}_2} + G_{\text{dB}_3} + \cdots + G_{\text{dB}_n}} \quad \text{dB}$$

Voltage Gains vs dB Levels

Comparing $A_v = \frac{V_o}{V_i}$ *to dB*

Voltage Gain, V_o/V_i	dB Level
0.5	−6
0.707	−3
1	0
2	6
10	20
40	32
100	40
1000	60
10,000	80
etc.	

GENERAL FREQUENCY CONSIDERATIONS

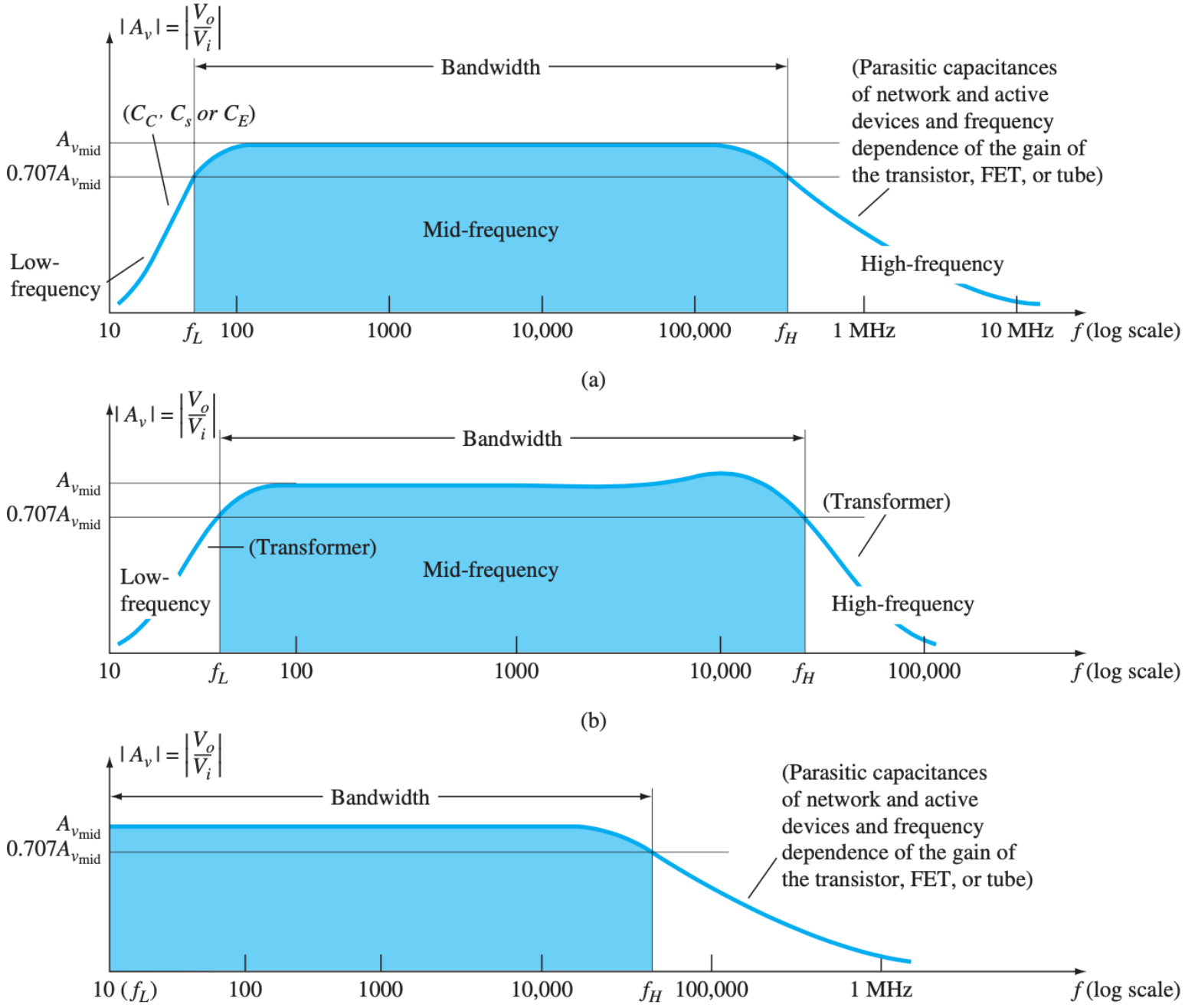


Figure Gain versus frequency:
(a) *RC*-coupled amplifiers;
(b) transformercoupledamplifiers;
(c) direct-coupled amplifiers.

0.707 where does it come from?

- When a voltage drops to 70.7%, the effective power it can produce into a resistive load is halved.
- So, the important thing to note is that a 50% power reduction is equivalent to the voltage reducing to √0.5 = 0.707106780 or 70.7% approximately.

$$P_{o_{\text{mid}}} = \frac{|V_o^2|}{R_o} = \frac{|A_{v_{\text{mid}}} V_i|^2}{R_o}$$

and at the half-power frequencies,

$$P_{o_{HPF}} = \frac{|0.707\, A_{v_{\text{mid}}} V_i|^2}{R_o} = 0.5\, \frac{|A_{v_{\text{mid}}} V_i|^2}{R_o}$$

$$P_{o_{\text{mid}}} = \frac{|V_o^2|}{R_o} = \frac{|A_{v_{\text{mid}}} V_i|^2}{R_o}$$

and at the half-power frequencies,

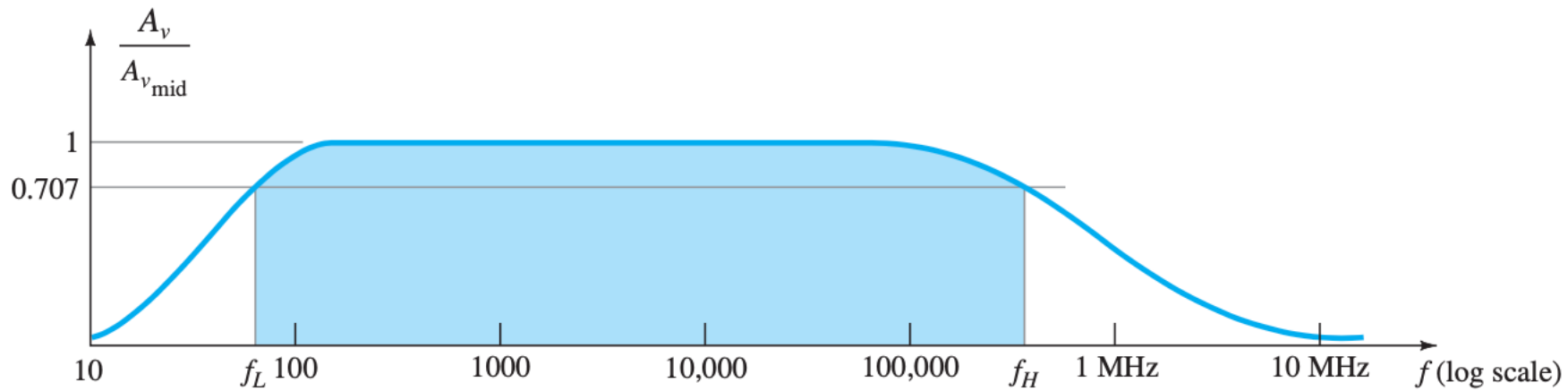
$$P_{o_{HPF}} = \frac{|0.707\, A_{v_{\text{mid}}} V_i|^2}{R_o} = 0.5\, \frac{|A_{v_{\text{mid}}} V_i|^2}{R_o}$$

and

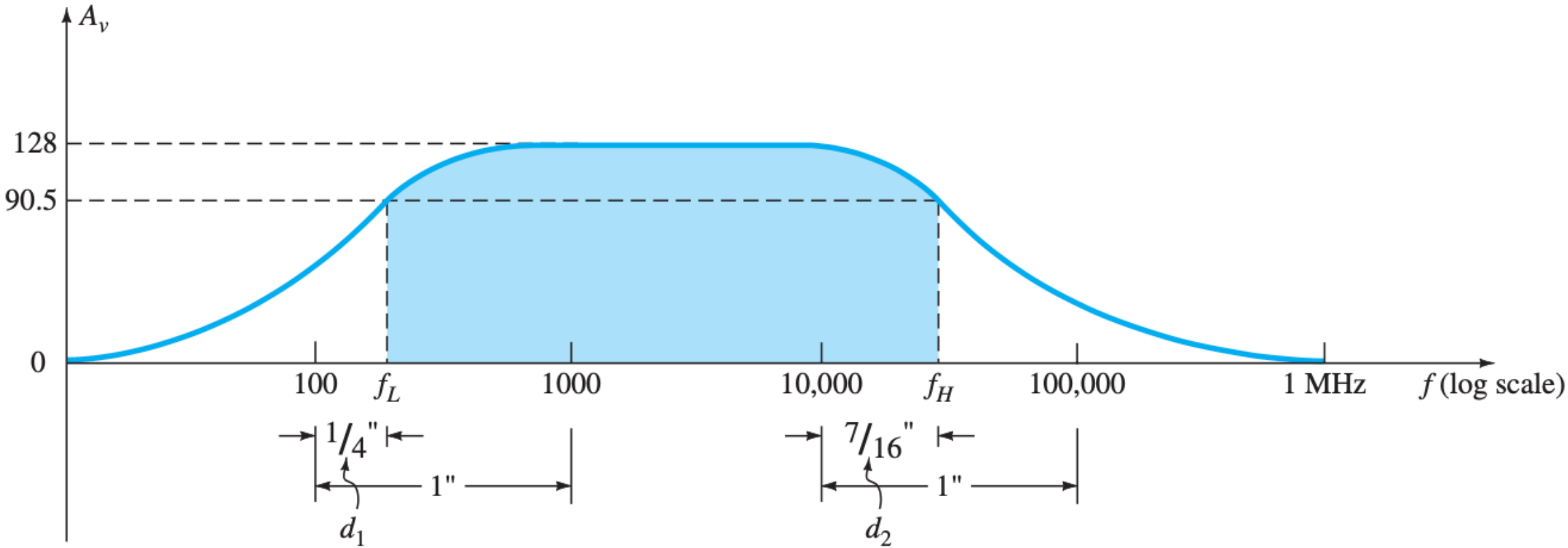
$$\boxed{P_{o_{HPF}} = 0.5\, P_{o_{\text{mid}}}}$$

The bandwidth (or passband) of each system is determined by f_H and f_L,

$$\boxed{\text{bandwidth (BW)} = f_H - f_L}$$



a. Find the cutoff frequencies f_L and f_H using the measurements provided.
b. Find the bandwidth of the response.
c. Sketch the normalized response.



Solution:

a. For f_L: $\dfrac{d_1}{d_2} = \dfrac{1/4''}{1''} = 0.25$

$10^{d_1/d_2} = 10^{0.25} = 1.7783$

Value $= 10^x \times 10^{d_1/d_2} = 10^2 \times 1.7783 =$ **177.83 Hz**

For f_H: $\dfrac{d_1}{d_2} = \dfrac{7/16''}{1''} = 0.438$

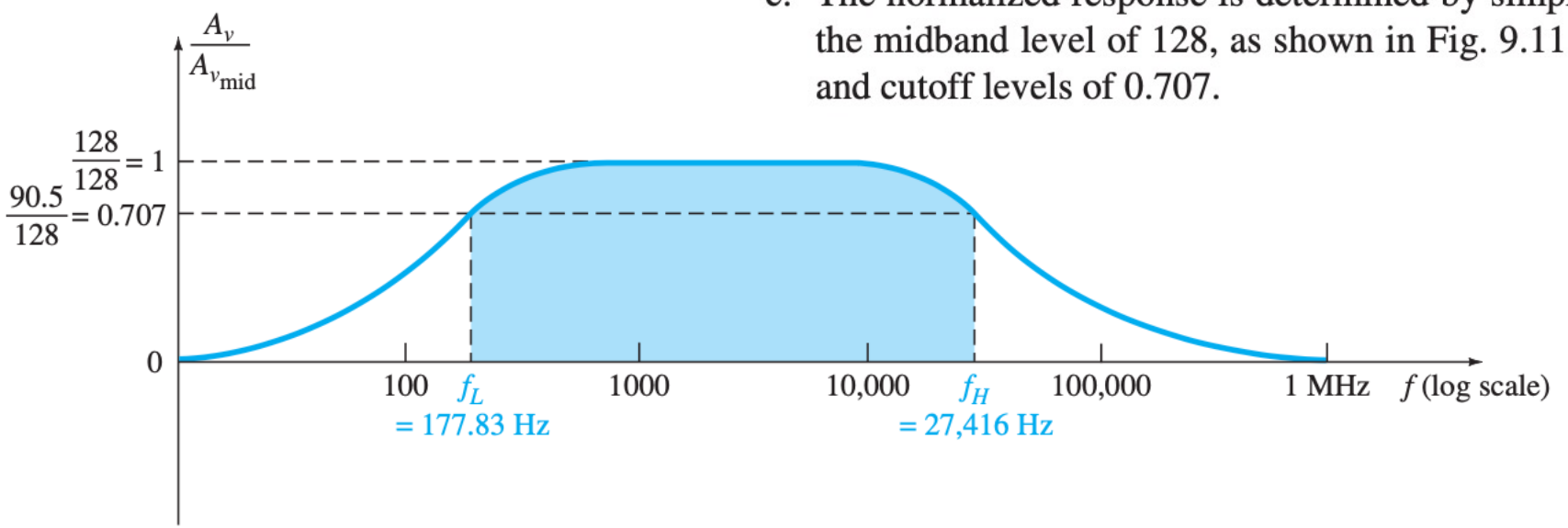
$10^{d_1/d_2} = 10^{0.438} = 2.7416$

Value $= 10^x \times 10^{d_1/d_2} = 10^4 \times 2.7416 =$ **27,416 Hz**

b. The bandwidth:

BW $= f_H - f_L = 27{,}416 \text{ Hz} - 177.83 \text{ Hz} \cong$ **27.24 KHz**

c. The normalized response is determined by simply dividing each level of Fig. 9.10 by the midband level of 128, as shown in Fig. 9.11. The result is a maximum value of 1 and cutoff levels of 0.707.



$$\left.\frac{A_v}{A_{v_{mid}}}\right|_{dB} = 20 \log_{10} \frac{A_v}{A_{v_{mid}}}$$

At midband frequencies, $20 \log_{10} 1 = 0$, and at the cutoff frequencies, $20 \log_{10} 1/\sqrt{2} = -3$ dB. Both values are clearly indicated in the resulting decibel plot of Fig. 9.12. The smaller the fraction ratio, the more negative is the decibel level.

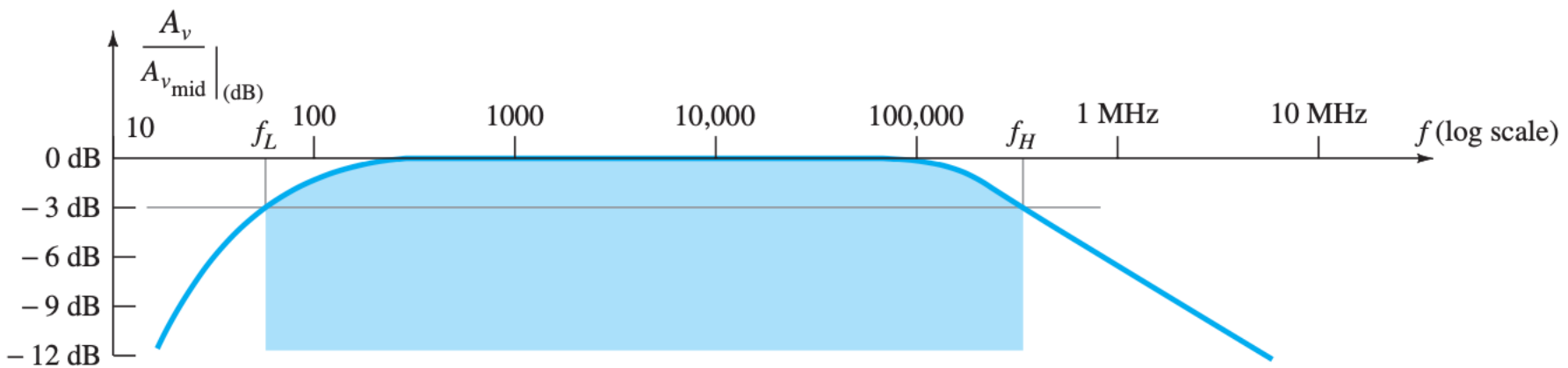


FIG. 9.12

Decibel plot of the normalized gain versus frequency plot of Fig. 9.9.

Most amplifiers introduce a 180° phase shift between input and output signals. This fact must now be expanded to indicate that this is the case only in the midband region. At low frequencies, there is a phase shift such that V_o lags V_i by an increased angle. At high frequencies, the phase shift drops below 180°. Figure 9.13 is a standard phase plot for an *RC*-coupled amplifier.

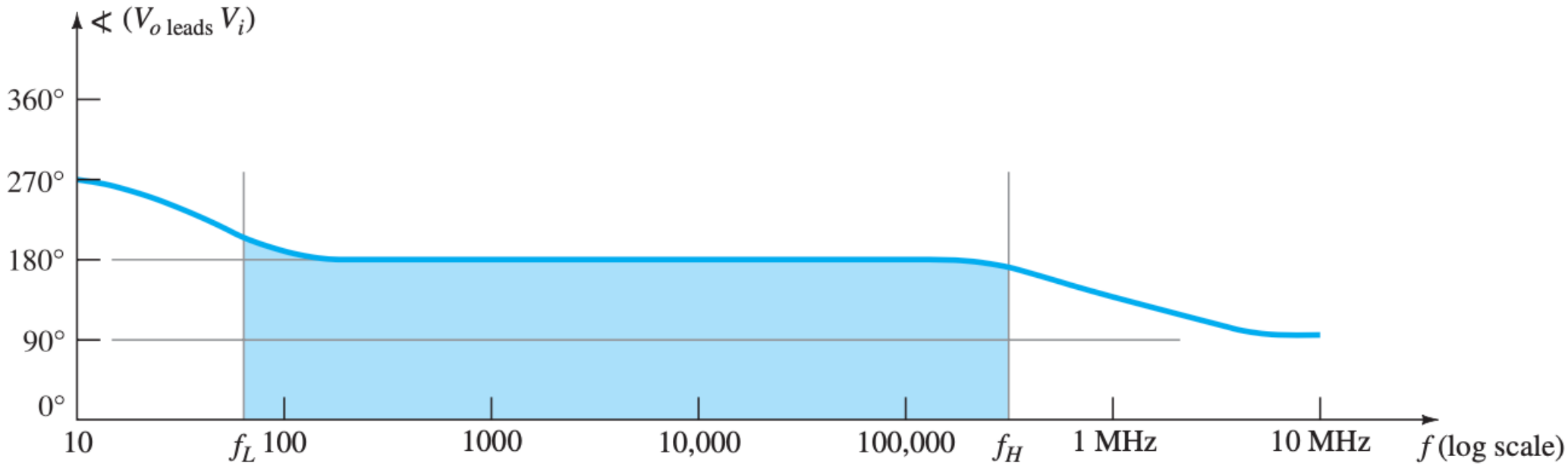
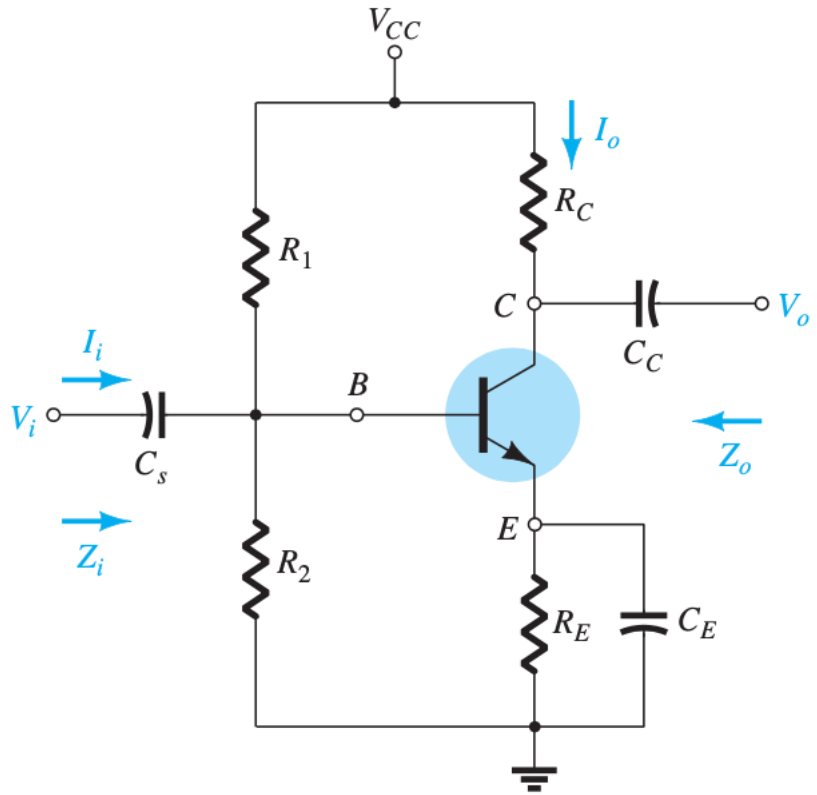


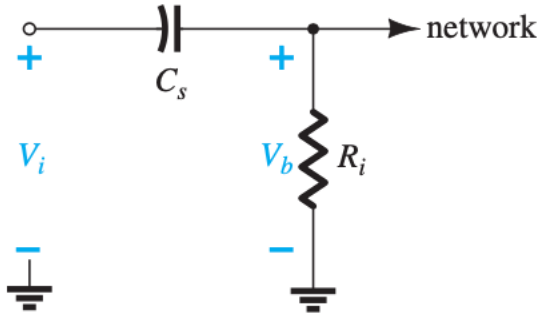
FIG. 9.13

Phase plot for an RC-coupled amplifier system.

LOW-FREQUENCY ANALYSIS



$$Z_i = R_i = R_1 \| R_2 \| \beta r_e$$



LOW-FREQUENCY ANALYSIS

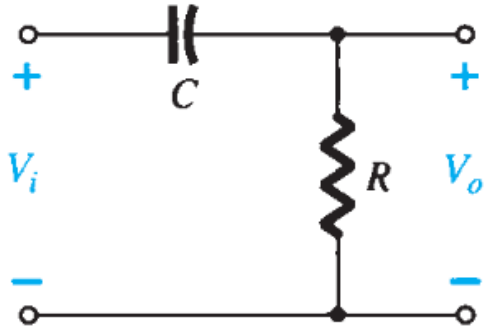


FIG. 9.14
RC combination that will define a low-cutoff frequency.

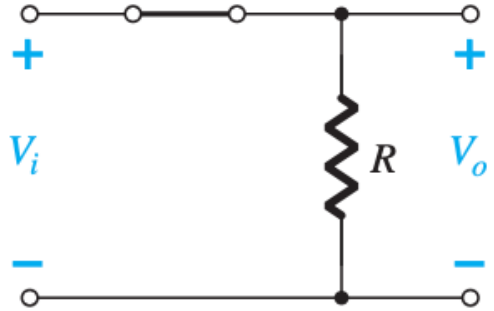


FIG. 9.17
RC circuit of Fig. 9.14 at very high frequencies.

$$X_C = \frac{1}{2\pi fC} \cong 0\ \Omega$$

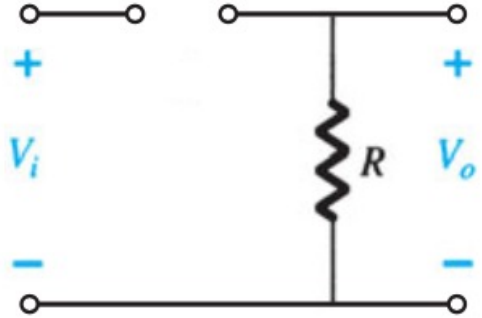


FIG. 9.18
RC circuit of Fig. 9.14 at $f = 0$ Hz.

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(0)C} = \infty\ \Omega$$

Between the two extremes, the ratio $A_v = V_o/V_i$ will vary as shown in Fig. 9.19. As the frequency increases, the capacitive reactance decreases, and more of the input voltage appears across the output terminals.

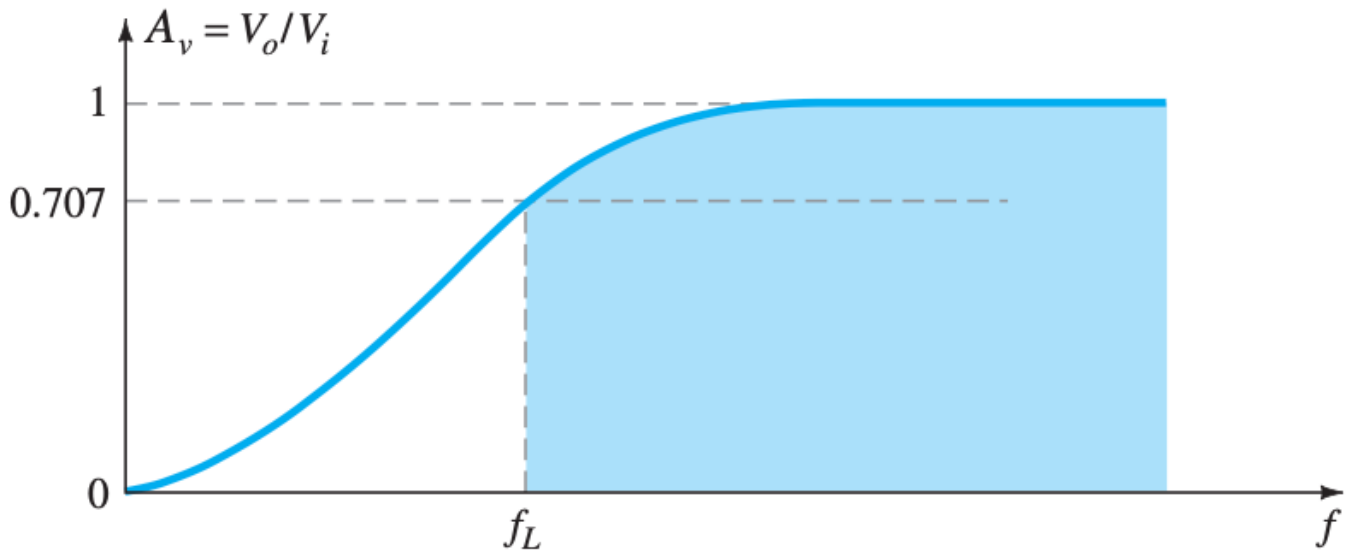


FIG. 9.19

Low-frequency response for the RC circuit of Fig. 9.14.

The output and input voltages are related by the voltage-divider rule in the following manner:

$$\mathbf{V}_o = \frac{\mathbf{R}\mathbf{V}_i}{\mathbf{R} + \mathbf{X}_C}$$

The magnitude of V_o is determined as follows:

$$V_o = \frac{RV_i}{\sqrt{R^2 + X_C^2}}$$

For the special case where $X_C = R$,

$$V_o = \frac{RV_i}{\sqrt{R^2 + X_C^2}} = \frac{RV_i}{\sqrt{R^2 + R^2}} = \frac{RV_i}{\sqrt{2R^2}} = \frac{RV_i}{\sqrt{2}R} = \frac{1}{\sqrt{2}}V_i$$

and

$$\boxed{|A_v| = \frac{V_o}{V_i} = \frac{1}{\sqrt{2}} = 0.707\big|_{X_C=R}} \qquad \textbf{(9.21)}$$

the level of which is indicated on Fig. 9.19. In other words, at the frequency for which $X_C = R$, the output will be 70.7% of the input for the network of Fig. 9.14.

The frequency at which this occurs is determined from

$$X_C = \frac{1}{2\pi f_L C} = R$$

and

$$\boxed{f_L = \frac{1}{2\pi RC}} \tag{9.22}$$

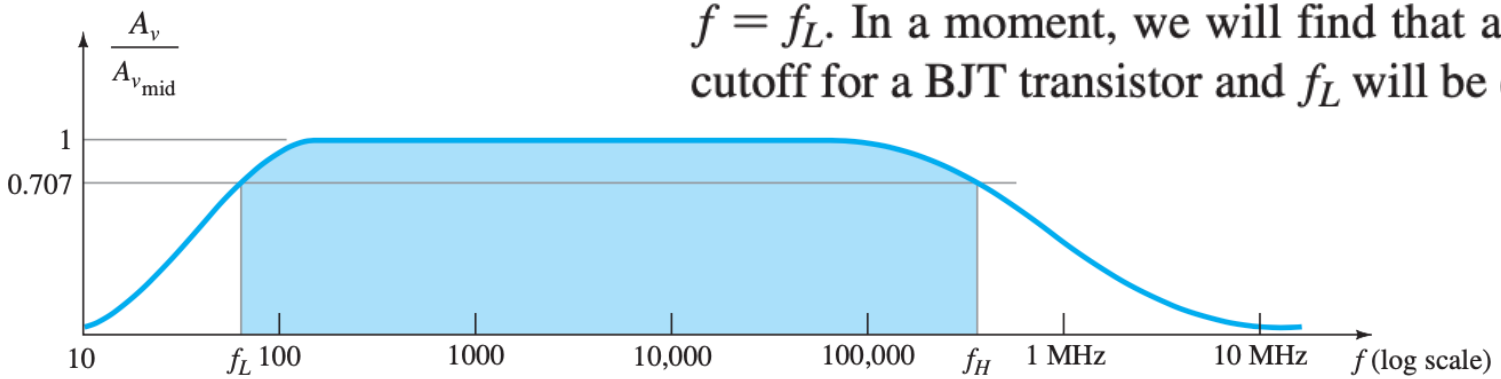
In terms of logs,

$$G_v = 20 \log_{10} A_v = 20 \log_{10} \frac{1}{\sqrt{2}} = -3 \text{ dB}$$

whereas at $A_v = V_o/V_i = 1$ or $V_o = V_i$ (the maximum value),

$$G_v = 20 \log_{10} 1 = 20(0) = 0 \text{ dB}$$

In Fig. 9.8, we recognize that there is a 3-dB drop in gain from the midband level when $f = f_L$. In a moment, we will find that an *RC* network will determine the low-frequency cutoff for a BJT transistor and f_L will be determined by Eq. (9.22).



If the gain equation is written as

$$A_v = \frac{V_o}{V_i} = \frac{R}{R - jX_C} = \frac{1}{1 - j(X_C/R)} = \frac{1}{1 - j(1/\omega CR)} = \frac{1}{1 - j(1/2\pi fCR)}$$

we obtain, using the frequency defined above,

$$\boxed{A_v = \frac{1}{1 - j(f_L/f)}} \tag{9.23}$$

In the magnitude and phase form,

$$A_v = \frac{V_o}{V_i} = \underbrace{\frac{1}{\sqrt{1 + (f_L/f)^2}}}_{\text{magnitude of } A_v} \underbrace{\angle \tan^{-1}(f_L/f)}_{\text{phase } \measuredangle \text{ by which } V_o \text{ leads } V_i} \tag{9.24}$$

For the magnitude when $f = f_L$,

$$|A_v| = \frac{1}{\sqrt{1 + (1)^2}} = \frac{1}{\sqrt{2}} = 0.707 \Rightarrow -3 \text{ dB}$$

In the logarithmic form, the gain in dB is

$$\boxed{A_{v(\text{dB})} = 20 \log_{10} \frac{1}{\sqrt{1 + (f_L/f)^2}}} \tag{9.25}$$

Expanding Eq. (9.25):

$$\begin{aligned} A_{v(\text{dB})} &= -20 \log_{10} \left[1 + \left(\frac{f_L}{f} \right)^2 \right]^{1/2} \\ &= -\left(\tfrac{1}{2}\right)(20) \log_{10} \left[1 + \left(\frac{f_L}{f} \right)^2 \right] \\ &= -10 \log_{10} \left[1 + \left(\frac{f_L}{f} \right)^2 \right] \end{aligned}$$

For frequencies where $f \ll f_L$ or $(f_L/f)^2 \gg 1$, the equation above can be approximated by

$$A_{v(\text{dB})} = -10 \log_{10} \left(\frac{f_L}{f} \right)^2$$

and finally,

$$\boxed{A_{v(\text{dB})} = -20 \log_{10} \frac{f_L}{f}} \quad f \ll f_L \tag{9.26}$$

Ignoring the condition $f \ll f_L$ for a moment, we find that a plot of Eq. (9.26) on a frequency log scale yields a result very useful for future decibel plots.

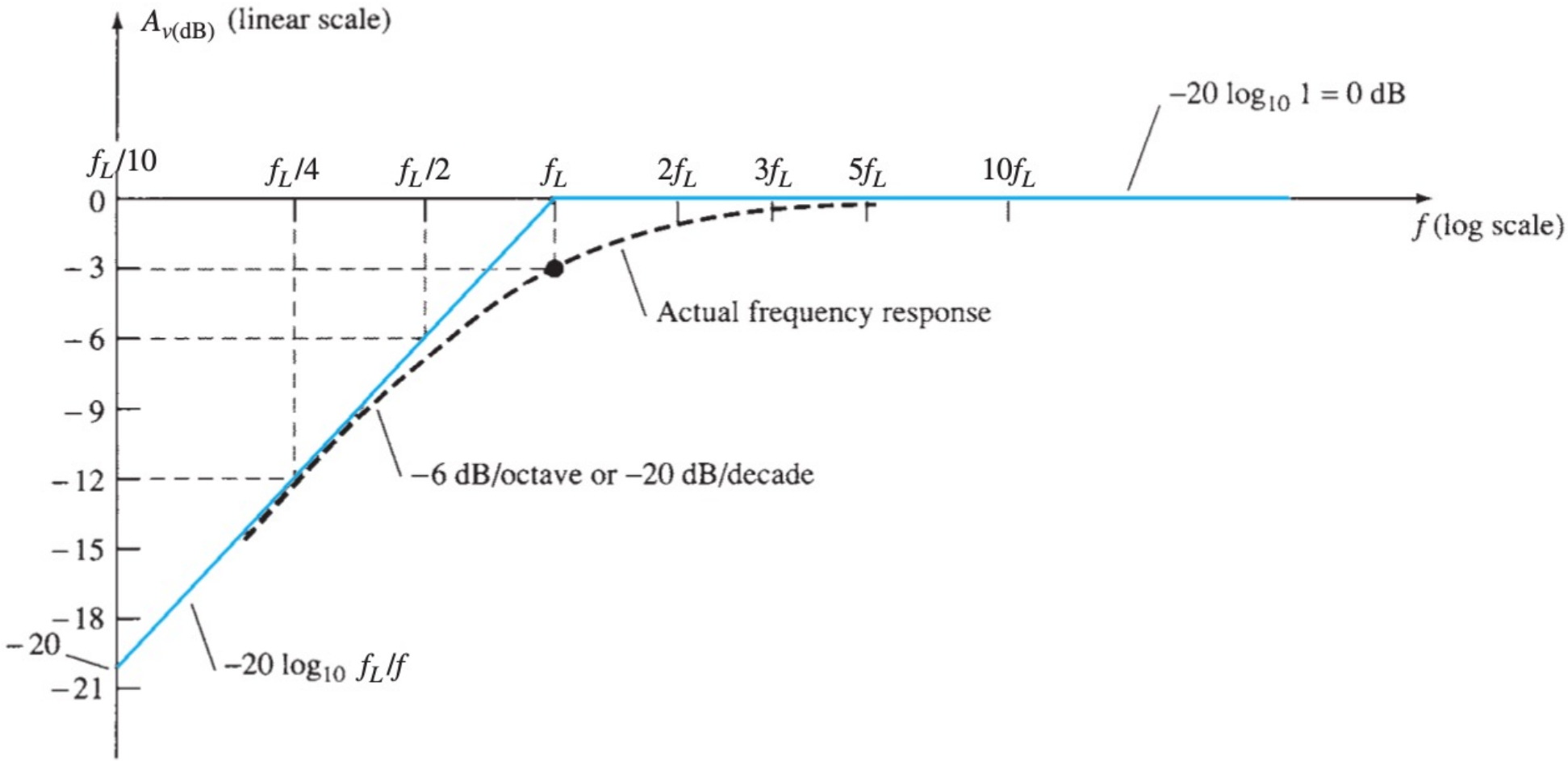
$$\text{At } f = f_L\text{:}\quad \frac{f_L}{f} = 1 \text{ and } -20\log_{10} 1 = 0 \text{ dB}$$

$$\text{At } f = \tfrac{1}{2}f_L\text{:}\quad \frac{f_L}{f} = 2 \text{ and } -20\log_{10} 2 \cong -6 \text{ dB}$$

$$\text{At } f = \tfrac{1}{4}f_L\text{:}\quad \frac{f_L}{f} = 4 \text{ and } -20\log_{10} 4 \cong -12 \text{ dB}$$

$$\text{At } f = \tfrac{1}{10}f_L\text{:}\quad \frac{f_L}{f} = 10 \text{ and } -20\log_{10} 10 = -20 \text{ dB}$$

The piecewise linear plot of the asymptotes and associated breakpoints is called a Bode plot of the magnitude versus frequency.



A change in frequency by a factor of two, equivalent to one octave, results in a 6-dB change in the ratio, as shown by the change in gain from $f_L/2$ to f_L.

As noted by the change in gain from $f_L/2$ to f_L:

For a 10:1 change in frequency, equivalent to one decade, there is a 20-dB change in the ratio, as demonstrated between the frequencies of $f_L/10$ and f_L.

Therefore, a decibel plot can easily be obtained for a function having the format of Eq. (9.26). First, simply find f_L from the circuit parameters and then sketch two asymptotes—one along the 0-dB line and the other drawn through f_L sloped at 6 dB/octave or 20 dB/decade. Then, find the 3-dB point corresponding to f_L and sketch the curve.

The gain at any frequency can be determined from the frequency plot in the following manner:

$$A_{v(\text{dB})} = 20 \log_{10} \frac{V_o}{V_i}$$

but

$$\frac{A_{v(\text{dB})}}{20} = \log_{10} \frac{V_o}{V_i}$$

and

$$\boxed{A_v = \frac{V_o}{V_i} = 10^{Av(\text{dB})/20}} \qquad \textbf{(9.27)}$$

For example, if $A_{v(\text{dB})} = -3$ dB,

$$A_v = \frac{V_o}{V_i} = 10^{(-3/20)} = 10^{(-0.15)} \cong 0.707 \qquad \text{as expected}$$

The quantity $10^{-0.15}$ is determined using the 10^x function found on most scientific calculators.

The phase angle of θ is determined from

$$\boxed{\theta = \tan^{-1} \frac{f_L}{f}} \qquad \textbf{(9.28)}$$

LOW-FREQUENCY RESPONSE BJT with RL

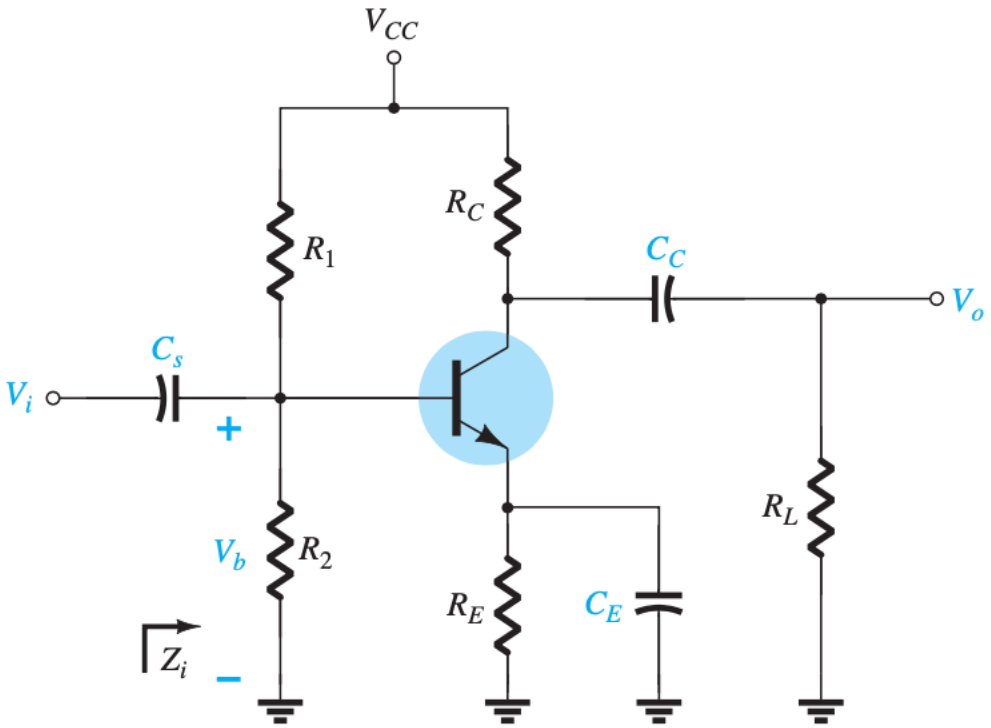


FIG. 9.25

Loaded BJT amplifier with capacitors that affect the low-frequency response.

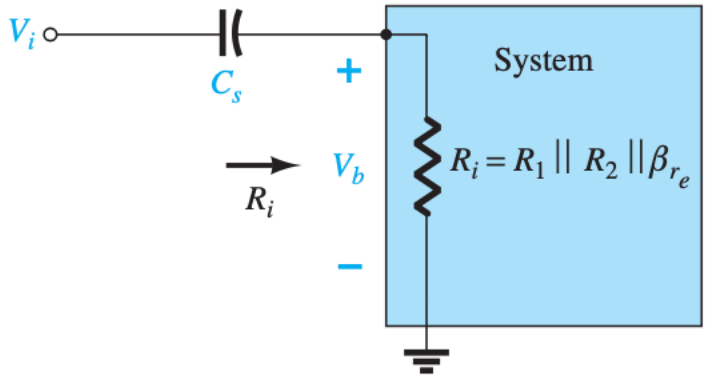


FIG. 9.26

Determining the effect of C_s on the low-frequency response.

Applying the voltage-divider rule:

$$\mathbf{V}_b = \frac{R_i \mathbf{V}_i}{R_i - jX_{C_s}} \tag{9.29}$$

The cutoff frequency defined by C_s can be determined by manipulating the above equation into a standard form or simply using the results of Section 9.6. As a verification of the results of Section 9.6 the manipulation process is defined in detail below. For future RC networks, the results of Section 9.6 will simply be applied.

Rewriting Eq. (9.29):

$$\frac{\mathbf{V}_b}{\mathbf{V}_i} = \frac{R_i}{R_i - jX_{C_s}} = \frac{1}{1 - j\dfrac{X_{C_s}}{R_i}}$$

The factor

$$\frac{X_{C_s}}{R_i} = \left(\frac{1}{2\pi f C_s}\right)\left(\frac{1}{R_i}\right) = \frac{1}{2\pi f R_i C_s}$$

Defining

$$f_{L_s} = \frac{1}{2\pi R_i C_s} \tag{9.30}$$

we have

$$\mathbf{A}_v = \frac{\mathbf{V}_b}{\mathbf{V}_i} = \frac{1}{1 - j(f_{L_s}/f)} \tag{9.31}$$

At f_{L_s} the voltage V_b will be 70.7% of the mid band value assuming C_s is the only capacitive element controlling the low-frequency response.

C_C Because the coupling capacitor is normally connected between the output of the active device and the applied load, the RC configuration that determines the low-cutoff frequency due to C_C appears in Fig. 9.27. The total series resistance is now $R_o + R_L$, and the cutoff frequency due to C_C is determined by

$$f_{L_C} = \frac{1}{2\pi(R_o + R_L)C_C} \tag{9.32}$$

Ignoring the effects of C_s and C_E, we find that the output voltage V_o will be 70.7% of its midband value at f_{L_C}. For the network of Fig. 9.25, the ac equivalent network for the output section with $V_i = 0$ V appears in Fig. 9.28. The resulting value for R_o in Eq. (9.32) is then simply

$$R_o = R_C \| r_o \tag{9.33}$$

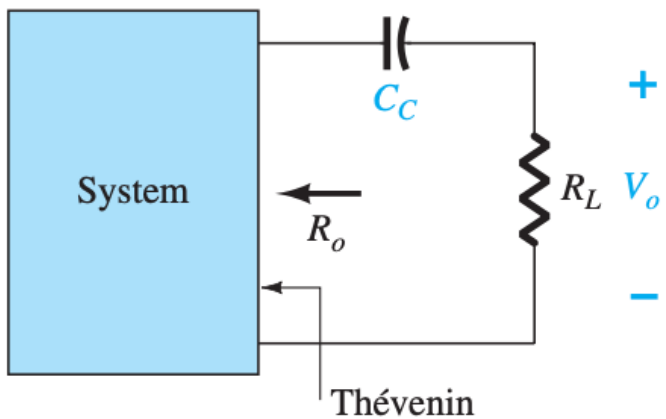


FIG. 9.27
Determining the effect of C_C on the low-frequency response.

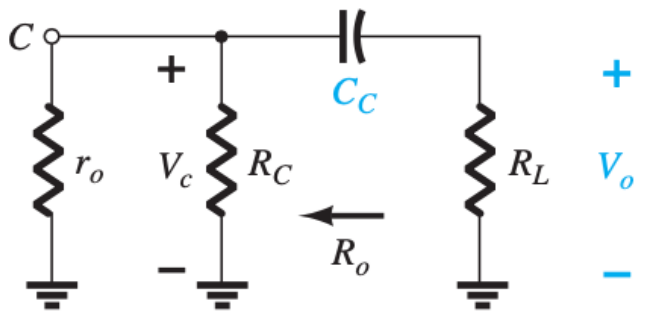


FIG. 9.28
Localized ac equivalent for C_C with $V_i = 0$ V.

C_E To determine f_{L_E}, the network "seen" by C_E must be determined as shown in Fig. 9.29. Once the level of R_e is established, the cutoff frequency due to C_E can be determined using the following equation:

$$f_{L_E} = \frac{1}{2\pi R_e C_E} \tag{9.34}$$

For the network of Fig. 9.25, the ac equivalent as "seen" by C_E appears in Fig. 9.30 as derived from Fig. 5.38. The value of R_e is therefore determined by

$$R_e = R_E \| \left(\frac{R_1 \| R_2}{\beta} + r_e \right) \tag{9.35}$$

The effect of C_E on the gain is best described in a quantitative manner by recalling that the gain for the configuration of Fig. 9.31 is given by

$$A_v = \frac{-R_C}{r_e + R_E}$$

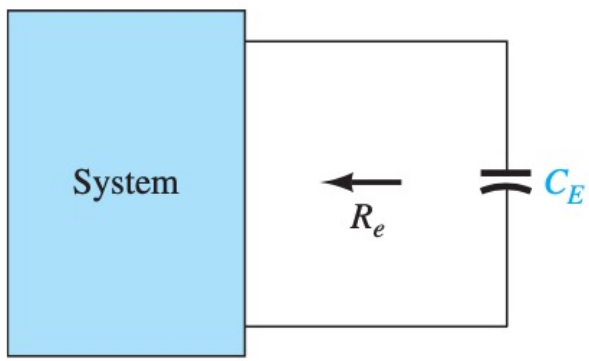


FIG. 9.29
Determining the effect of C_E on the low-frequency response.

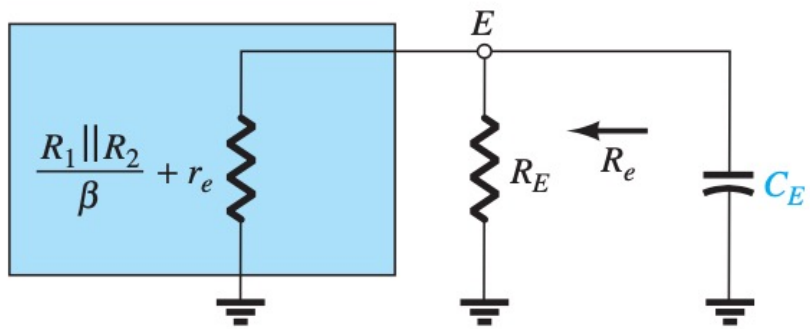


FIG. 9.30
Localized ac equivalent of C_E.

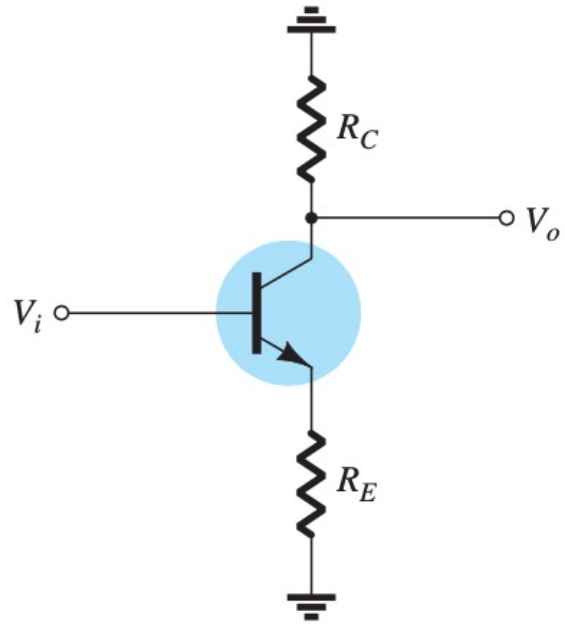


FIG. 9.31
Network employed to describe the effect of C_E on the amplifier gain.

The maximum gain is obviously available where R_E is 0 Ω. At low frequencies, with the bypass capacitor C_E in its "open-circuit" equivalent state, all of R_E appears in the gain equation above, resulting in the minimum gain. As the frequency increases, the reactance of the capacitor C_E will decrease, reducing the parallel impedance of R_E and C_E until the resistor R_E is effectively "shorted out" by C_E. The result is a maximum or midband gain determined by $A_v = -R_C/r_e$. At f_{L_E} the gain will be 3 dB below the midband value determined with R_E "shorted out."

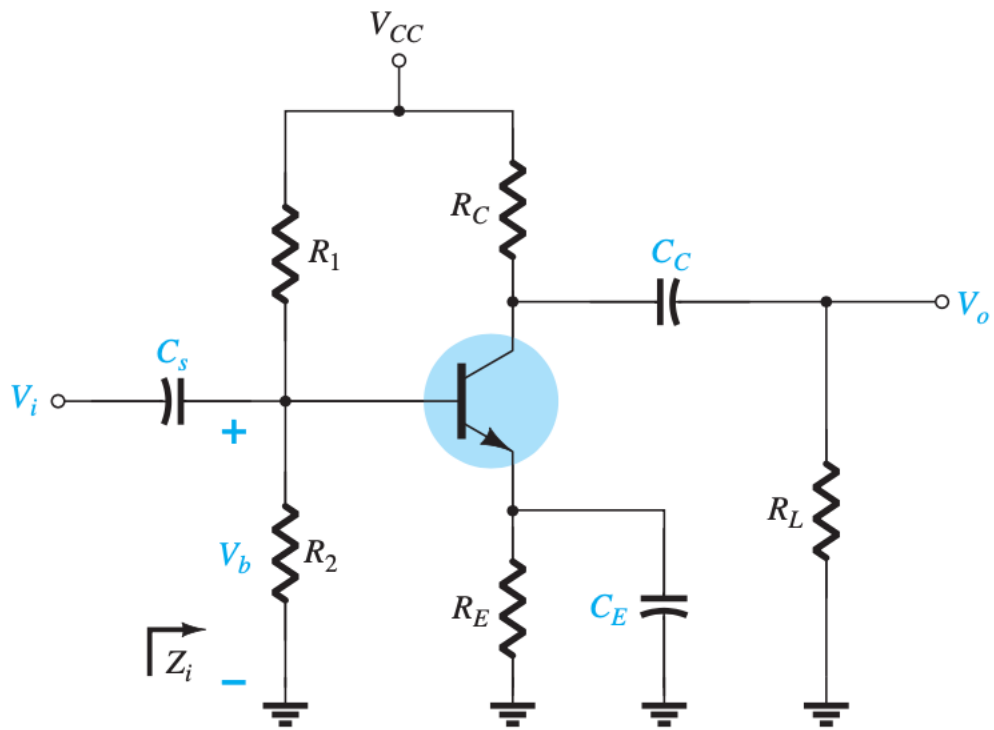
EXAMPLE 9.11 Determine the cutoff frequencies for the network of Fig. 9.25 using the following parameters:

$C_s = 10\,\mu\text{F}, \quad C_E = 20\,\mu\text{F}, \quad C_C = 1\,\mu\text{F}$

$R_1 = 40\,\text{k}\Omega, \quad R_2 = 10\,\text{k}\Omega, \quad R_E = 2\,\text{k}\Omega, \quad R_C = 4\,\text{k}\Omega,$

$R_L = 2.2\,\text{k}\Omega$

$\beta = 100, \quad r_o = \infty\Omega, \quad V_{CC} = 20\,\text{V}$



Step 1

Solution: To determine r_e for dc conditions, we first apply the test equation:

$$\beta R_E = (100)(2\,\text{k}\Omega) = 200\,\text{k}\Omega \gg 10R_2 = 100\,\text{k}\Omega$$

Since satisfied the dc base voltage is determined by

$$V_B \cong \frac{R_2 V_{CC}}{R_2 + R_1} = \frac{10\,\text{k}\Omega(20\,\text{V})}{10\,\text{k}\Omega + 40\,\text{k}\Omega} = \frac{200\,\text{V}}{50} = 4\,\text{V}$$

with

$$I_E = \frac{V_E}{R_E} = \frac{4\,\text{V} - 0.7\,\text{V}}{2\,\text{k}\Omega} = \frac{3.3\,\text{V}}{2\,\text{k}\Omega} = 1.65\,\text{mA}$$

so that

$$r_e = \frac{26\,\text{mV}}{1.65\,\text{mA}} \cong \mathbf{15.76\ \Omega}$$

and

$$\beta r_e = 100(15.76\ \Omega) = 1576\ \Omega = \mathbf{1.576\,k\Omega}$$

Step 2 (1/2)

Midband Gain

$$A_v = \frac{V_o}{V_i} = \frac{-R_C \| R_L}{r_e} = -\frac{(4\,\text{k}\Omega)\|(2.2\,\text{k}\Omega)}{15.76\,\Omega} \cong -90$$

C_s

$$R_i = R_1 \| R_2 \| \beta r_e = 40\,\text{k}\Omega \| 10\,\text{k}\Omega \| 1.576\,\text{k}\Omega \cong 1.32\,\text{k}\Omega$$

$$f_{L_S} = \frac{1}{2\pi R_i C_s} = \frac{1}{(6.28)(1.32\,\text{k}\Omega)(10\,\mu\text{F})}$$

$$f_{L_S} \cong \mathbf{12.06\ Hz}$$

C_C

$$f_{L_C} = \frac{1}{2\pi(R_o + R_L)C_C} \quad \text{with} \quad R_o = R_C \| r_o \cong R_C$$

$$= \frac{1}{(6.28)(4\,\text{k}\Omega + 2.2\,\text{k}\Omega)(1\,\mu\text{F})}$$

$$\cong \mathbf{25.68\ Hz}$$

Step 2 (2/2)

C_E

$$R_e = R_E \| \left(\frac{R_1 \| R_2}{\beta} + r_e \right)$$
$$= 2\,\text{k}\Omega \| \left(\frac{40\,\text{k}\Omega \| 10\,\text{k}\Omega}{100} + 15.76\,\Omega \right)$$
$$= 2\,\text{k}\Omega \| \left(\frac{8\,\text{k}\Omega}{100} + 15.76\,\Omega \right)$$
$$= 2\,\text{k}\Omega \| (80\,\Omega + 15.76\,\Omega)$$
$$= 2\,\text{k}\Omega \| 95.76\,\Omega$$
$$= 91.38\,\Omega$$

$$f_{L_E} = \frac{1}{2\pi R_e C_E} = \frac{1}{(6.28)(91.38\,\Omega)(20\,\mu\text{F})} = \frac{10^6}{11{,}477.73} \cong \mathbf{87.13\ Hz}$$

Since $f_{L_E} \gg f_{L_C}$ or f_{L_S} the bypass capacitor C_E is determining the lower cutoff frequency of the amplifier.

The piecewise linear plot of the asymptotes and associated breakpoints is called a Bode plot of the magnitude versus frequency.

